

Statement — Student Marks and Grading System

Project Submission Document

1. Problem Statement

Calculating student totals, percentages, grades, and pass/fail results manually can take time and may lead to calculation errors. Managing student records and finding a particular student's details can also be inconvenient.

The **Student Marks and Grading System** provides a simple menu-driven Python program to record student information, process marks, calculate results, and manage student records.

2. Scope of the Project

The project covers basic student result and record management using Python. It accepts a student's name, roll number, and marks in five subjects:

- Maths
- English
- Hindi
- Physics
- Chemistry

Each subject is marked out of 100, giving a maximum total of 500 marks.

The system can:

- Add new student records.
- Validate marks entered by the user.
- Calculate total marks and percentage.
- Assign grades from A+ to F.
- Determine PASS or FAIL based on marks in every subject.
- Display all student records.
- Search for a student using a roll number.
- Update an existing student record.
- Delete a student record.
- Display a class summary containing total students, pass/fail counts, class average, highest percentage, and top student's name.

The project stores records in memory while the program is running. Permanent file/database storage is outside the current scope.

3. Target Users

The intended users are:

- Students learning Python programming and basic data processing.
- Teachers or faculty members who want a simple demonstration of student result management.
- Users who need a basic command-line system for calculating and viewing student marks and grades.

4. High-Level Features

Student Record Management

- Add student name, roll number, and marks.
- Prevent duplicate roll numbers.
- Display all stored student records.
- Search for a student by roll number.
- Update student details and marks.
- Delete a student record.

Marks Validation

- Accept marks only from 0 to 100.
- Handle non-numeric input and request valid input again.

Result Calculation

- Calculate total marks out of 500.
- Calculate percentage automatically.
- Assign grades according to the defined grading scale.
- Mark a student PASS only when every subject has at least 40 marks.

Grading Scale

Percentage	Grade
90 and above	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
40 to below 50	E
Below 40	F

|---|---|

| 90 and above | A+ |

| 80 to below 90 | A |

| 70 to below 80 | B |

| 60 to below 70 | C |

| 50 to below 60 | D |

| 40 to below 50 | E |

| Below 40 | F |

Class Summary

The class summary displays:

- Total number of students.
- Number of passed students.
- Number of failed students.
- Class average percentage.
- Highest percentage.
- Name of the top student.

5. Technologies / Tools Used

- **Python** — main programming language.
- **Google Colab** — development/execution environment.
- **GitHub** — project source-code and documentation repository.
- Python functions, loops, conditional statements, lists, dictionaries, input validation, and basic calculations.

6. Project Details

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Program: Integrated M.Tech – Artificial Intelligence

Year: 1st Year

Course: Python Essential

College: VIT Bhopal University

Academic Year: 2026–27

7. Example Result

For a student scoring 90 marks in each of the five subjects:

- Total: 450 / 500
- Percentage: 90.0%
- Grade: A+
- Result: PASS

8. Limitations

- Student records are stored only in memory while the program is running.
- Records are not automatically saved to a file or database.
- The project currently uses a fixed set of five subjects.
- The grading rules are fixed in the program.

9. Future Enhancements

Possible future improvements include:

- Permanent storage using files or a database.
- A graphical user interface.

- PDF report-card generation.
- Excel export of student results.
- Configurable subjects and grading rules.
- Login and access control.

10. Conclusion

The Student Marks and Grading System demonstrates how Python can be used to solve a practical student-result management problem. The project combines functions, loops, conditional statements, lists, dictionaries, input validation, and calculations to make student mark processing and record management simpler.

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